From: North Dakota University System (NDUS)

To: Office of Science and Technology Policy (OSTP)

Subject: NDUS Response to the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Date: March 14, 2025

Introduction

The development and implementation of artificial intelligence (AI) technologies present a transformative opportunity for the United States. However, it is crucial to ensure that the benefits of AI are equitably distributed across all states, including rural states like North Dakota. This response outlines the rationale for prioritizing equitable AI development and highlights specific initiatives in North Dakota that can serve as models for other rural states and for the country.

Rationale for the Broad-based Development of AI Capabilities across the United States

A national AI Action Plan should truly be nationwide, leaving no state or region behind.

1. **Economic Revitalization**: Rural states often face economic challenges, including declining industries and outmigration. AI can drive economic revitalization by creating new job opportunities, attracting tech companies, and fostering innovation in local industries such as agriculture, energy, and manufacturing.
2. **Educational Opportunities**: Access to AI education and training can empower residents of rural states with the skills needed for high-demand jobs. This can help bridge the digital divide and ensure that rural populations are not left behind in the AI-driven economy.
3. **National Security**: Ensuring that all regions have robust AI capabilities is critical for national security. Rural states can play a vital role in cybersecurity and defense, contributing to a more resilient national infrastructure.
4. **Innovation and Diversity**: Diverse perspectives from different regions can drive innovation in AI. Rural states bring unique challenges and opportunities that can lead to novel AI applications and solutions.

5. **NSF Engines and EDA Tech Hub Examples**:
    a. The National Science Foundation's Regional Innovations Engines program is specifically designed to broaden the innovation base across the United States, taking advantage of talent that's found in every state.
    b. The Department of Commerce's Economic Development Administration's Tech Hubs program strengthens U.S. economic and national security with investments in regions across the country whose assets and resources have the potential to become globally competitive in the technologies and industries of the future.
    c. These are examples of national plans for broad based development that the AI Action Plan could employ with great effect.
6. **Rural Representation in AI Governance Bodies**. If the AI Action Plan contemplates creation of Federal AI Governance or Advisory entities, rural states should be afforded equitable representation from the outset by design.

Rationale for Equitable Federal and Private Investment in AI Capabilities

Unleashing innovation and investment shouldn't be focused on existing technology geographies. All states have resources that can contribute to the effort. Federal and private investment incentives should be geographically equitable.

1. **Balanced National Growth**: Equitable investment in AI capabilities ensures balanced growth across the nation. By supporting AI initiatives in rural states, the federal government and private sector partners can prevent economic disparities and promote uniform development[1].
2. **Maximizing Talent Utilization**: Rural states have untapped talent pools that can significantly contribute to advancements in AI. Equitable investment can help harness this potential, leading to a more inclusive and comprehensive AI workforce[2].
3. **Enhanced Innovation Ecosystem**: Federal support for AI in rural areas can foster a more robust and diverse innovation ecosystem. This can lead to the development of AI solutions tailored to the unique needs and challenges of different regions[3].
4. **Strengthening National Security**: Distributed AI capabilities across various states enhance national security by reducing the risk of centralized vulnerabilities. Rural states can play a crucial role in developing and implementing AI-driven security measures[1].

Examples of Current Federal Investment in AI Capabilities

Federal investment to date has been important for the development of AI. This investment should continue with an eye to expanding its reach to include rural states.

1. **National Science Foundation (NSF)**: The NSF has significantly increased its investment in AI research, with a focus on advancing AI technologies and applications. For FY25, the NSF requested $2.05 billion, with a substantial portion dedicated to AI-focused research[1].
2. **Department of Defense (DOD)**: The DOD, through the Defense Advanced Research Projects Agency (DARPA), continues to invest heavily in AI research and development. These investments aim to enhance national security and develop cutting-edge AI technologies[2].
3. **Department of Energy (DOE)**: The DOE supports AI research through its national laboratories and research programs. These investments focus on leveraging AI for energy efficiency, grid management, and scientific research[1].
4. **National Institutes of Health (NIH)**: The NIH invests in AI to advance medical research and healthcare. AI applications in this sector include disease diagnosis, personalized medicine, and drug discovery[2].
5. **Network and Information Technology Research and Development (NITRD) Program**: The NITRD program, established under the High Performance Computing Act of 1991, continues to provide foundational support for AI and IT research across multiple federal agencies[1].

North Dakota AI and Cyber Initiatives

North Dakota is a thought leader in how to use AI to advance local, state, tribal, regional, and national priorities. We're an example of how all states, including rural states, can contribute to the successful development of AI in the country. A sampling of examples includes:

1. **Mountain-Plains University Innovation Alliance**: This Alliance was created in 2020 to harness the physical resources and intellectual capacity of the public universities in Idaho, Montana, North Dakota, South Dakota, and Wyoming. This existing Alliance is ready to contribute to AI research and applications on a regional basis, allowing our region to achieve the scale essential to AI initiatives.
2. **Cyber Land Grant University Act**: Chancellor Mark Hagerott of the North Dakota University System has proposed the Cyber Land Grant University Act, modeled after the Morrill Act of 1862. This initiative aims to establish a national

network of digital-cyber universities to democratize access to digital skills and address the concentration of technological expertise in urban areas.

3. **Cybersecurity Centers of Excellence**: North Dakota universities, including North Dakota State University (NDSU), the University of North Dakota (UND), Minot State University (MiSU), and Bismarck State College (BSC), have been accredited as Centers of Academic Excellence in Cyber Defense by the NSA and DHS. These designations recognize North Dakota's commitment to cybersecurity education and research and our ability to contribute to national AI initiatives.
4. **AI Institute for Teaching and Learning**: Valley City State University (VCSU) has proposed the establishment of an AI Institute for Teaching and Learning. This institute aims to integrate AI into educational practices, enhancing teaching methodologies and preparing students for the AI-driven future. This builds on VCSU's long-standing leadership and innovation in the development of the education workforce.
5. **National Center for the Application of AI in Technical Education**: The North Dakota State College of Science (NDSCS) has proposed the creation of a National Center for the Application of AI in Technical Education. This center would focus on applying AI technologies to technical education, ensuring that students acquire practical, job-ready skills. This proposal leverages NDSCS' record of success in training and educating the skilled workforce our nation needs.
6. **National Security Crossroads Initiative**: North Dakota is in a unique position to boost and sustain AI-driven Unmanned Aerial Systems (UAS) and counter UAS (C-UAS) capabilities for national security. Geographically, this "crossroads" could link up Grand Sky UAS test airspace near the City of Grand Forks and Grand Forks Air Force Base, a test site north of Highway 2, a test site at Camp Grafton Training Center and the airspace surrounding Minot Air Force Base. Higher education will have a key role in UAS/C-UAS research, development, education, and training resources. Six NDUS institutions in the Crossroads can contribute to this initiative: Minot State Univer-sity, Dakota College at Bottineau, Lake Region State College, Mayville State University, the Univer-sity of North Dakota, and North Dakota State University
7. **Autonomous Systems Research and Development**: The goal of the research and development of AI-driven autonomous systems at North Dakota's two R1 research institutions is to create new autonomous systems through multidisciplinary research and lead development of world-changing autonomous policies, with the twin goals of driving a vibrant, diverse, and sustainable U.S. economy and contributing capabilities for national defense. The University of North Dakota's Research Institute for Autonomous Systems is a global leader in unmanned autonomous systems research, application, and policy development. North Dakota State University's research through its Agricultural Experiment

Station, its Ag Technology Project, and its Upper Great Plains Transportation Institute aims to place AI-driven autonomy at the forefront of its agricultural research agenda.

Conclusion

Ensuring that rural states benefit equally from the development and implementation of AI is not only a matter of equity but also a strategic imperative for the nation's economic, educational, and national security interests. The initiatives in North Dakota provide a blueprint for how rural states can leverage AI to drive growth and innovation. We urge the OSTP to consider these points and support policies that promote equitable AI development across all regions, including the investments made by the Federal Government.

Contact Information

Mark Hagerott, Ph.D.
Chancellor
North Dakota University System
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[1]: Federal AI and IT Research and Development Spending Analysis

[2]: R&D in IT and Artificial Intelligence: Resources and Focus Areas

[3]: National Artificial Intelligence Initiative (NAII)

References

[1] Federal AI and IT Research and Development Spending Analysis
[2] R&D in IT and Artificial Intelligence: Resources and Focus Areas
[3] Behind the FTC's 6(b) Report on Large AI Partnerships & Investments